Ain Shams Engineering Journal (2014) 5, 169–175

Ain Shams University

Ain Shams Engineering Journal

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MECHANICAL ENGINEERING

Influence of thermal radiation, viscous dissipation and Hall current on MHD convection flow over a stretched vertical flat plate

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Received 16 April 2013; revised 24 August 2013; accepted 30 August 2013
Available online 5 October 2013

KEYWORDS

Thermal radiation;
Viscous dissipation;
Hall current effect;
MHD;
Stretched flat plate

Abstract A boundary layer analysis has been presented to study the effects of thermal radiation, viscous dissipation, and Hall current effects on the hydromagnetic convection flow of an electrically conducting, viscous, incompressible fluid past over a stretching vertical flat plate. The governing partial differential equations are non-dimensionalized and transformed into a system of nonlinear ordinary differential similarity equations, in a single independent variable. The resulting coupled nonlinear equations are solved under appropriate transformed boundary conditions using the Runge–Kutta fourth-order method along with the shooting method. The effects of the radiation parameter, magnetic parameter, Hall parameter, viscous dissipation parameter and the buoyancy parameter on the velocity profiles, the cross flow velocity profiles and the temperature profiles are presented graphically and discussed.

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1. Introduction

The field of boundary layer flow problem over a stretching sheet has many industrial applications such as polymer sheet or filament extrusion from a dye or long thread between feed roll or wind-up roll, glass fiber and paper production, drawing of plastic films, liquid films in condensation process. Due to the high applicability of this problem in such industrial phenomena, it has attracted the attentions of many researchers and one of the pioneering studies has been performed by Sakiadis [1]. The study of stretching surfaces and the several combinations of additional effects on the stretching problems are important in many practical applications because the production of sheeting material arises in a number of industrial manufacturing processes and includes both metal and polymer sheets. In the manufacture of the latter, the material is in a molten phase when thrust through an extrusion die and then cools and solidifies some distance away from the die before arriving at the collecting stage. The quality of the resulting sheeting material, as well as the cost of production, is affected by the speed of collection and the heat transfer rate, and a knowledge of the flow properties of the ambient fluid is clearly desirable (see Banks and Zaturska [2]). On the other hand, it should be pointed out that the very important practical prob-

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http://dx.doi.org/10.1016/j.asej.2013.08.003

Nomenclature

a,c	constants
B_0	the strength of the imposed magnetic field
C_{fx}	skin friction coefficient in x-direction
C_{fz}	skin friction coefficient in z-direction
Ec	Eckert number
e	electric charge
f	dimensionless stream function
f'	dimensionless velocity
g	acceleration due to gravity
Gr_x	local Grashof number
h	cross flow velocity
H_0	external magnetic field
k	thermal conductivity of the fluid
k_e	mean absorption coefficient
m	Hall parameter
m_e	the mass of an electron
M	magnetic parameter
n_e	electron number density
Pr	Prandtl number
q_r	radiative heat flux
R	radiation parameter
Re_x	local Reynolds number
T	fluid temperature
T_e	electron collision time
T_w	surface temperature
T_∞	ambient temperature
u, v, w	velocity components along the x, y and z directions, respectively
$u_w(x)$	velocity of the stretching plate
x, y, z	Cartesian coordinates along the stretching surface, normal to it, and transverse to the xy plane, respectively

Greek symbols

α	thermal diffusivity
β	thermal expansion coefficient
λ	constant buoyancy parameter
σ_s	Stefan-Boltzmann constant
θ	dimensionless temperature
ν	kinematic viscosity
μ	dynamic viscosity
μ_e	magnetic permeability
ρ	fluid density
τ_w	wall shear stress

Subscripts

w	condition at the surface
∞	ambient condition

lems of the thermal processing of sheet-like materials which is a necessary operation in the production of paper, linoleum, polymeric sheets, roofing shingles, insulating materials, fine-fiber matts are described in the excellent papers by Sparrow and Abraham [3] and Abraham and Sparrow [4]. Other very important applications of the stretching sheets are described in the papers by Lakshmisha et al. [5].

Magnetohydrodynamics (MHD) is the study of the interaction of conducting fluids with electromagnetic phenomena. The flow of an electrically conducting fluid in the presence of a magnetic field is of importance in various areas of technology and engineering such as MHD power generation, MHD flow meters, MHD pumps etc. The viscous flow due to stretching boundary is important in extrusion processes where sheet material is pulled out of an orifice with increasing velocity. It is worth mentioning that Sakiadis [6] was the first to study the boundary layer flow on a continuously moving solid surface. Crane [7] extended this idea for the steady two-dimensional boundary layer flow due to a stretching sheet whose velocity varies linearly with the distance from a fixed point in the sheet. Recently, the study of magnetohydrodynamics (MHD) become important in engineering applications, such as in designing cooling system with liquid metals, MHD generator and other devices in the petroleum industry. For ionized gases, the conventional MHD is not valid under the strong electric field. In an ionized gas, the density is low and the magnetic field is very strong, the conductivity normal to the magnetic field is reduced due to the free spiralling of electrons and ions about the magnetic lines of force before suffering collisions, and also a current is induced in the direction normal to both electric and magnetic fields. This phenomenon is also known as the Hall effect (Sato [8], Sutton and Sherman [9], Pop [10], Raptis and Ram [11], Hossain and Rashid [12]).

The case of non-isothermal stretching flat plate in the presence of a transverse magnetic field where the Hall effect has been neglected was considered by Ezzat et al. [13] and Zakaria [14]. Yih [15] studied the effect of free convection on MHD heat and mass transfer of a continuously moving permeable vertical surface. The study of MHD free convection flow along a vertical surface with the effects of Hall current over a porous plate has been considered by Hossain [16], Hossain and Mohammad [17]. Also, Salem and Abd El-Aziz [18] have studied the effects of Hall current and chemical reaction on the hydromagnetic flow of a stretching vertical surface. Recently, Ghosh et al. [19] have studied the Hall effects in a parallel plate channel, while Abd El-Aziz [20] has analyzed the effects of Hall currents on the flow and heat transfer of an electrically conducting fluid over an unsteady stretching surface in the presence of a strong magnetic field.

In the context of space technology and in the processes involving high temperatures, the effects of radiation are of vital importance. Recent developments in hypersonic flights, missile re-entry, rocket combustion chambers, power plants for inter planetary flight and gas cooled nuclear reactors, have focused attention on thermal radiation as a mode of energy transfer, and emphasized the need for improved understanding of radiative transfer in these processes. The interaction of radiation with laminar free convection heat transfer from a vertical plate was investigated by Cess [21] for an absorbing, emitting fluid in the optically thick region, using the singular perturbation technique. Arpaci [22] considered a similar problem in both the optically thin and optically thick regions and

used the approximate integral technique and first-order profiles to solve the energy equation. Gnaneswara Reddy and Bhaskar Reddy [23] studied the radiation and mass transfer effects on unsteady MHD free convection flow past a vertical porous plate with viscous dissipation by using finite element method. Recently, Gnaneswara Reddy and Bhaskar Reddy [24] investigated mass transfer and heat generation effects on MHD free convection flow past an inclined vertical surface in a porous medium. MHD boundary-layer flow over a stretching surface with internal heat generation or absorption was studied by Basiri Parsa et al. [25]. Gnaneswara Reddy [26] analyzed thermophoresis, viscous dissipation and joule heating effects on steady MHD heat and mass transfer flow over an inclined radiative isothermal permeable surface with variable thermal conductivity.

The objective of the present paper is to study the effects of Joule heating on steady MHD mixed convection boundary layer flow over a stretched vertical flat plate in the presence of thermal radiation and viscous dissipation. The partial differential equations are reduced to similarity or nonlinear ordinary differential equations which are solved numerically. The effects of various governing parameters on the velocity, cross flow velocity, temperature, skin-friction coefficient and Nusselt number are shown in figures and tables and discussed in detail.

2. Mathematical analysis

Consider the steady mixed convection flow of an incompressible, viscous and electrically conducting fluid past a stretching flat plate in the vertical direction with a velocity proportional to the distance from the fixed origin O of a stationary frame of reference (x, y, z). The frame of reference (x, y, z) is chosen such that the x-axis is along the direction of motion of the surface, the y-axis is normal to the surface and the z-axis is transverse to the xy-plane. We consider that an external constant magnetic field H_0 is applied in the positive y-direction. It is also assumed that the surface of the sheet has a variable temperature.

$T_w(x)$, while T_∞, $T_w(x) > T_\infty$ where corresponds to a heated plate and $T_w(x) < T_\infty$ corresponds to a cooled plate.

Taking Hall effects into account and assuming that the electron pressure gradient, the ion slip and the thermo-electric effects are neglected, the generalized Ohm's law can be written as (Cowling [27])

$$\mathbf{j} = \sigma\left[\mathbf{E} + \mu_e v \times \mathbf{H} - \frac{\mu_e}{en_e}\mathbf{j} \times \mathbf{H}\right] \tag{1}$$

where the electron pressure gradient and ion-slip effects are neglected. In (1), n_e and e stand for the electron number density and the electric charge, respectively and the electrical conductivity, σ is given by

$$\sigma = e^2 n_e T_e / m_e \tag{2}$$

where T_e and m_e denote the electron collision time and the mass of an electron, respectively. The effect of Hall current gives rise to a force in the z-direction resulting in a cross-flow in this direction and thus the flow becomes three-dimensional. Using the boundary layer variables with the observation that the physical variables do not depend on the coordinate z.

Under the above stated assumptions and usual Boussinesq's approximation, the governing equations relevant to the problem are:

Continuity equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{3}$$

Momentum equations

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = v\frac{\partial^2 u}{\partial y^2} + g\beta(T - T_\infty) - \frac{\sigma B_0^2}{\rho(1+m^2)}(u + mw) \tag{4}$$

$$u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} = v\frac{\partial^2 w}{\partial y^2} + \frac{\sigma B_0^2}{\rho(1+m^2)}(mu - w) \tag{5}$$

Energy equation

$$u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} = \frac{k}{\rho c_p}\frac{\partial^2 T}{\partial y^2} + \frac{\mu}{\rho c_p}\left[\left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2\right] - \frac{1}{\rho c_p} \times \frac{\partial q_r}{\partial y} \tag{6}$$

The boundary conditions are

$$u = u_w(x), \quad v = w = 0, \quad T = T_w(x) at y = 0 \tag{7}$$

$$u \to 0, \quad w \to 0, \quad T \to T_\infty \ as \ y \to \infty$$

By using the Rosseland approximation Brewster [28], the radiative heat flux q_r is given by

$$q_r = -\frac{4\sigma_s}{3k_e}\frac{\partial T^4}{\partial y} \tag{8}$$

where σ_s is the Stefan–Boltzmann constant and k_e is the mean absorption coefficient. It should be noted that by using the Rosseland approximation, the present analysis is limited to optically thick fluids. If the temperature differences within the flow are sufficiently small, then Eq (8) can be linearized by expanding T^4 into the Taylor series about T_∞, which after neglecting higher order terms takes the form

$$T^4 \cong 4T_\infty^3 T - 3T_\infty^4 \tag{9}$$

In view of Eqs (8) and (9), Eq (6) reduces to

$$u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} = \alpha(1+R)\frac{\partial^2 T}{\partial y^2} + \frac{\mu}{\rho c_p}\left[\left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2\right] \tag{10}$$

where $\alpha = k/\rho c_p$ is the thermal diffusivity and $R = 16\sigma^* T_\infty^3/3kk^*$ is the radiation parameter.

In order for (3)–(5) and (10) admit a similarity solution, we assume that $u_w(x)$ and $T_w(x)$ vary linearly with the variable x as

$$u_w(x) = cx, \quad T_w(x) = T_\infty + ax \tag{11}$$

where $c(>0)$ and a are constants. We notice that for $a > 0$ $(T_w(x) > T_\infty)$, the plate is heated and for $a < 0$ $(T_w(x) < T_\infty)$ the plate is cooled.

We look for a similarity solution of (3)–(6) of the form

$$u = cxf'(\eta), \quad v = -(cv)^{1/2}f(\eta), \quad w = cxh(\eta), \quad \theta(\eta) = \frac{T - T_\infty}{T_w - T_\infty},$$
$$\eta = (c/v)^{1/2}y,$$
$$M = \frac{\sigma B_0^2}{c\rho}, \quad Gr_x = \frac{g\beta(T_w - T_\infty)x^3}{v^2}, \quad \lambda = \frac{Gr_x}{\mathrm{Re}_x^2}, \quad \mathrm{Pr} = \frac{v}{\alpha},$$
$$Ec = \frac{v^2}{c_p(T_w - T_\infty)} \tag{12}$$

where primes denote differentiation with respect to η.

By substituting (12) into (4), (5), and (10), we obtain the following ordinary differential equations

$$f''' + ff'' - f'^2 - \frac{M}{1+m^2}(f' + mh) + \lambda\theta = 0 \tag{13}$$

$$H'' + fh' - f'h + \frac{M}{1+m^2}(mf' - h) = 0 \tag{14}$$

$$(1+R)\theta'' + \Pr[f\theta' - f'\theta + Ec(f''^2 + h''^2)] = 0 \tag{15}$$

The corresponding boundary conditions take the form

$$\begin{aligned} &f = 0, \quad f' = 1, \quad h = 0, \quad \theta = 1 \ at \ \eta = 0 \\ &f' \to 0, \quad h \to 0, \quad \theta \to 0 \ as \ \eta \to \infty \end{aligned} \tag{16}$$

where Pr is the Prandtl number, M is the magnetic parameter, m is the Hall parameter, λ is the constant buoyancy or mixed convection parameter, Gr_x is the local Grashof number and $\mathrm{Re}_x = \frac{u_w(x)}{\nu}$ is the local Reynolds number and Ec is the Eckert number.

It should be mentioned that $\lambda > 0$ corresponds to the assisting flow (heated plate) while $\lambda < 0$ corresponds to the opposing flow (cooled plate) and $\lambda = 0$ corresponds to the forced convection flow.

The quantities of physical interest are the skin friction coefficients C_{f_x} and C_{f_z} as well as the Nusselt number Nu, which are defined as

$$C_{f_x} = \frac{\tau_{w_x}^2}{\rho u_w}, \quad C_{f_z} = \frac{\tau_{w_z}^2}{\rho u_w}, \quad Nu = \frac{xq_w}{k(T_w - T_\infty)} \tag{17}$$

where τ_{w_x} and τ_{w_z} are the wall shear stresses in the directions of x and z, respectively, while q_w is the heat flux from the surface of the flat plate, which are given by

$$\begin{aligned} \tau_{w_x} &= \mu\left(\frac{\partial u}{\partial y}\right)_{y=0}, \quad \tau_{w_z} = \mu\left(\frac{\partial w}{\partial y}\right)_{y=0}, \quad q_w \\ &= -k\left(\frac{\partial T}{\partial y}\right)_{y=0} \end{aligned} \tag{18}$$

with k being the thermal conductivity of the fluid. Using (12), we get

$$\begin{aligned} C_{f_x}\mathrm{Re}_x^{1/2} &= f''(0), \quad C_{f_z}\mathrm{Re}_x^{1/2} = h'(0), \quad Nu\mathrm{Re}_x^{1/2} \\ &= -\theta'(0) \end{aligned} \tag{19}$$

3. Results and discussion

The set of nonlinear ordinary differential Eqs. (13)–(15) with boundary conditions (16) have been solved by using the Runge–Kutta fourth order along with Shooting method. First of all, higher order non-linear differential Eqs. (13)–(15) are converted into simultaneous linear differential equations of first order and they are further transformed into initial value problem by applying the shooting technique (Jain et al. [29]). The resultant initial value problem is solved by employing Runge–Kutta fourth order technique. The step size $\Delta\eta = 0.01$ is used to obtain the numerical solution with five decimal place accuracy as the criterion of convergence. However, we have considered only the assisting flow ($\lambda > 0$) and forced convection flow ($\lambda = 0$) cases.

In order to assess the accuracy of the numerical method, results for $-\theta'(0)$ in the absence of Hall current ($m = 0$), magnetic field ($M = 0$), constant buoyancy parameter ($\lambda = 0$), thermal radiation ($R = 0$) and viscous dissipation ($R = 0$); were compared with those of Yih [15] and Ali et al. [30] and found to be in good agreement. These comparisons are shown in Table 1. The effects of various governing parameters on the skin-friction coefficient $f''(0)$ and local Nusselt number $-\theta'(0)$ are shown in Table 2. It is observed that as the Hall current parameter m or the constant buoyancy or mixed convection parameter λ increases the magnitude of the skin friction coefficient $f''(0)$ and local Nusselt $-\theta'(0)$ number increases. It is found that as the magnetic parameter M increases the magnitude of the skin friction coefficient $f''(0)$ and local Nusselt $-\theta'(0)$ number decreases. It is noticed that as the radiation parameter R or Eckert number Ec increases, the magnitude of the skin friction coefficient $f''(0)$ also increases while the local Nusselt number, $-\theta'(0)$ decreases.

To analyze the results, numerical computation has been carried out for variations in the governing parameters such as the Prandtl number Pr, magnetic field parameter M, Hall parameter m, constant buoyancy or mixed convection parameter λ and Eckert number Ec.

In the present study following default parameter values are adopted for computations: $\Pr = 0.71$, $M = 1.0$, $m = 1.0$, $\lambda = 1.0$, $R = 1.0$ and $Ec = 0.01$. All graphs therefore correspond to these values unless specifically indicated on the appropriate graph.

In order to assess the accuracy of our computed results, the present result has been compared with Ali et al. [30] for different values of Pr on velocity, cross flow velocity and temperature profiles shown Figs. 1–3 with $R = Ec = 0.0$. It is observed that the agreements with the solution of temperature profiles are excellent.

Figs. 4–6 show the effects of the Hall parameter m and constant buoyancy or mixed convection parameter λ on the velocity f', the cross flow velocity h and the temperature θ profiles, respectively. The effect of increasing m has increased the velocity profiles f' and the cross flow velocity profiles h, however it decreases the temperature profiles θ. It is observed that the effect of the Hall parameter m on the temperature is small and that the magnetic and Hall parameters have opposite

Table 1 Comparison of results for the heat transfer rate $-\theta'(0)$ with $m = 0$, $\lambda = 0$, $M = 0$, $R = 0$ and $Ec = 0$.

Pr	Yih [15]	Ali el al. [30]	Present results
0.71	0.8086	0.8086	0.80864
1	1.0000	1.0000	1.0001
3	1.9237	1.9237	1.9230
10	3.7207	3.7208	3.72028

Table 2 Numerical values of the skin-friction coefficient and Nusselt number for $Pr = 0.71$.

m	λ	M	R	Ec	$f''(0)$	$-\theta(0)$
1.0	1.0	1.0	1.0	0.01	−0.707122	0.584872
2.0	1.0	1.0	1.0	0.01	−0.554359	0.605207
1.0	3.0	1.0	1.0	0.01	0.185339	0.681148
1.0	1.0	2.0	1.0	0.01	−0.972345	0.54508
1.0	1.0	1.0	3.0	0.01	−0.653399	0.417914
1.0	1.0	1.0	1.0	0.06	−0.70602	0.57886

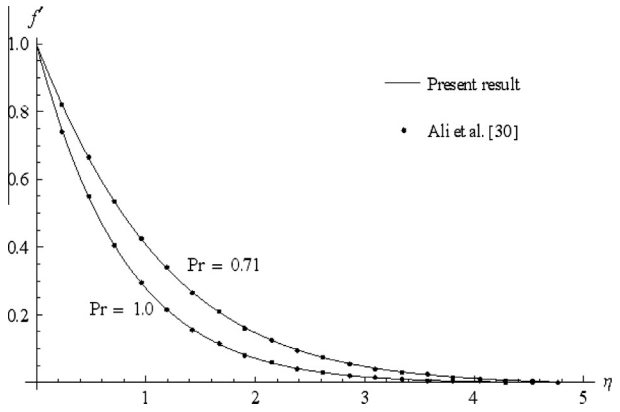


Figure 1 Comparison of velocity profiles.

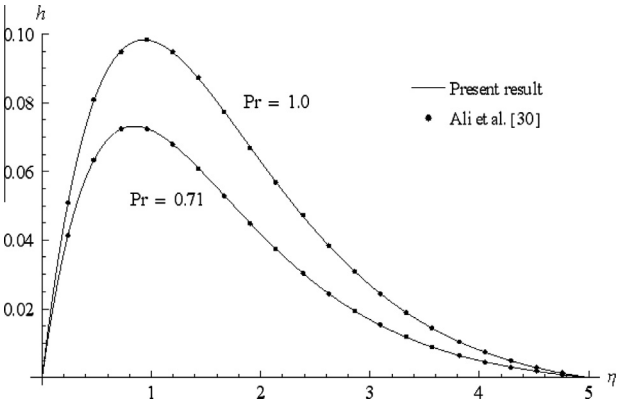


Figure 2 Comparison of cross flow velocity profiles.

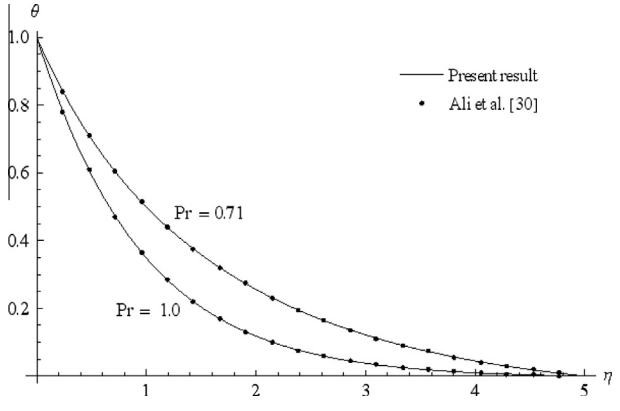


Figure 3 Comparison of temperature profiles.

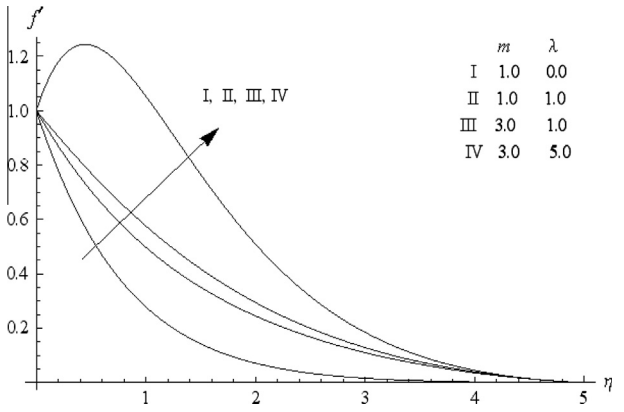


Figure 4 Velocity profiles for different values of m, λ.

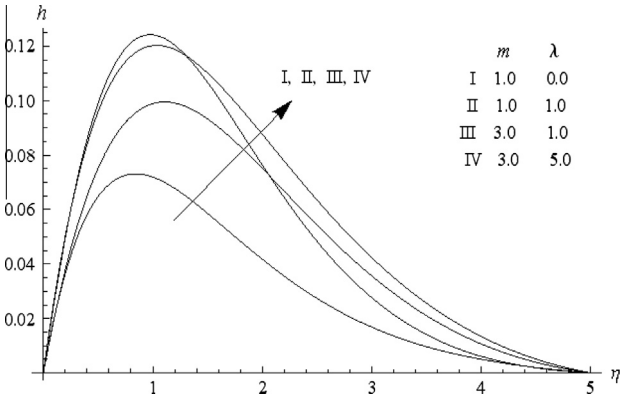


Figure 5 Cross flow velocity profiles for different values of m, λ.

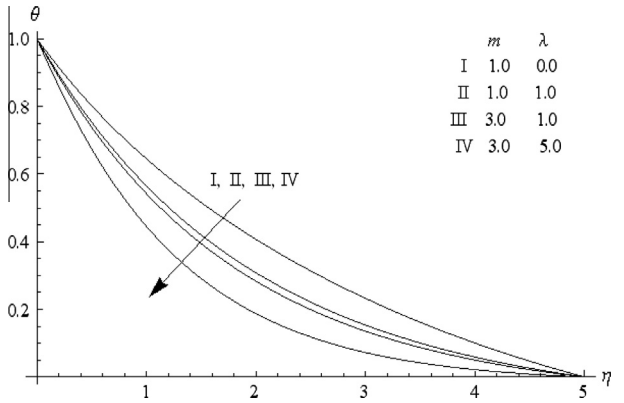


Figure 6 Temperature profiles for different values of m, λ.

effects of the velocity and temperature profiles. The curve for the case $\lambda = 0$ (forced convection flow) is also included in each figure in order to see the changes from forced convection to mixed convection flows. When m and M are fixed at $m = 1.0$ and $M = 1.0$, the maximum value of the velocity and the cross flow velocity profiles increase with increasing the parameter λ, while the temperature profiles decrease with increasing λ. It is shown that the thickness of the velocity boundary layer is much smaller than the thickness of the thermal boundary layer due to the small value of Pr, and this is a well-known fact for low Prandtl number fluids.

Figs. 7 and 8 show the effects of magnetic parameter, M on the velocity f' and the temperature θ profiles, respectively. The curve for the case $M = 0.0$ (magnetic field is absent) is also included in each figure in order to see the changes. As expected, the velocity f' and the temperature profiles θ increase with the increase in M. When M increases, this will also increase the Lorentz force which opposes the flow and leads to enhanced deceleration of the velocity profiles. The Prandtl number defines the ratio of momentum diffusivity to thermal diffusivity. It is noticed that an increase in the Prandtl number results a

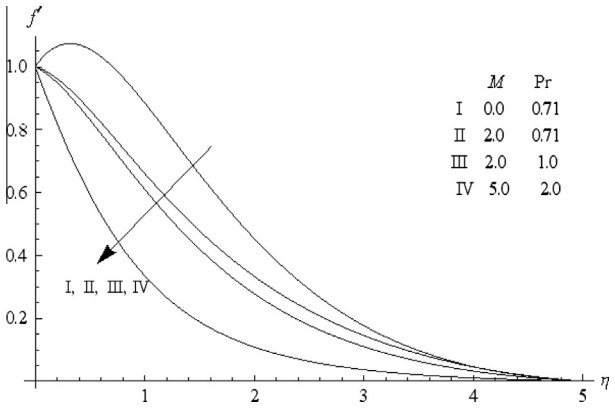


Figure 7 Velocity profiles for different values of M, Pr.

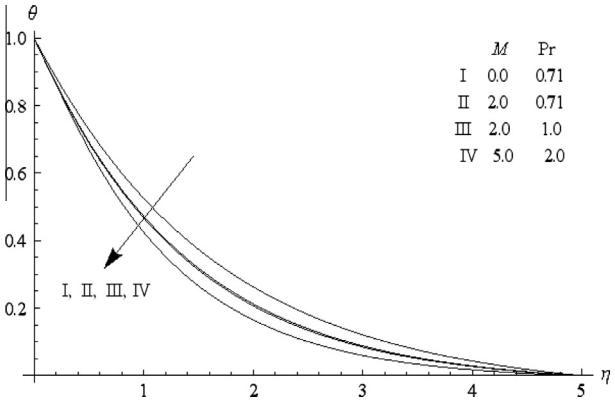


Figure 8 Temperature profiles for different values of M, Pr.

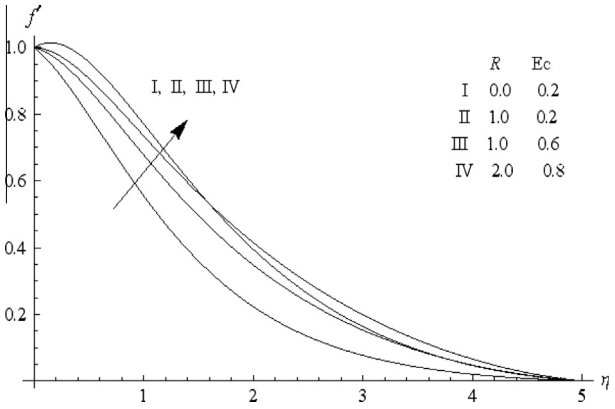


Figure 9 Velocity profiles for different values of R, Ec.

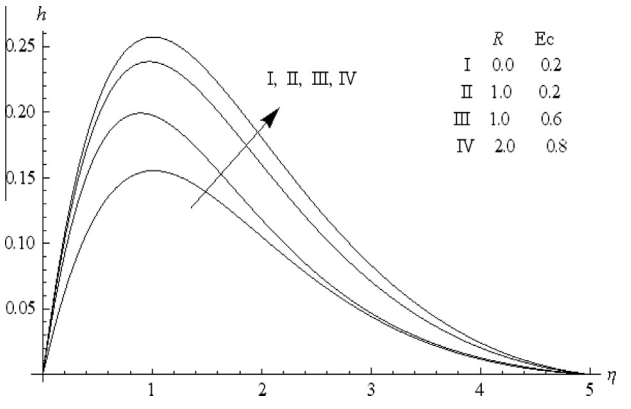


Figure 10 Cross flow velocity profiles for different values of R, Ec.

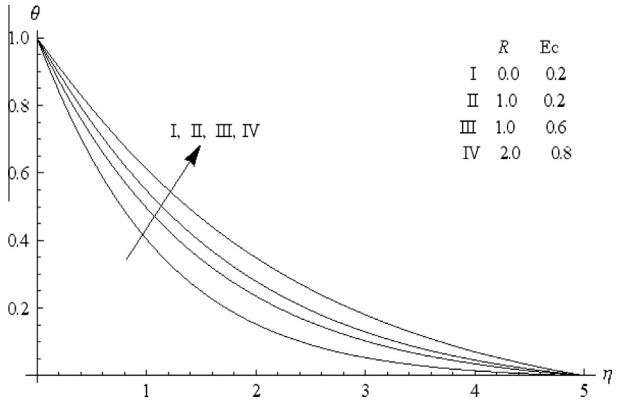


Figure 11 Temperature profiles for different values of R, Ec.

decrease of the thermal boundary layer thickness and in general lower average temperature within the boundary layer. The reason is that smaller values of Pr are equivalent to increasing the thermal conductivities, and therefore heat is able to diffuse away from the heated plate more rapidly than for higher values of Pr. Hence in the case of smaller Prandtl numbers as the boundary layer is thicker and the rate of heat transfer is reduced.

The influence of the radiation parameter R and viscous dissipation parameter i.e., the Eckert number Ec on the velocity f', the cross flow velocity h and the temperature θ profiles are shown in Figs. 9–11 respectively. The radiation parameter R defines the relative contribution of conduction heat transfer to thermal radiation transfer. It is obvious that an increase in the radiation parameter results in increasing velocity, cross flow velocity and temperature within the boundary layer. The Eckert number Ec expresses the relationship between the kinetic energy in the flow and the enthalpy. It embodies the conversion of kinetic energy into internal energy by work done against the viscous fluid stresses. Greater viscous dissipative heat causes a rise in the temperature, as well as the velocity and cross flow velocity. This behavior is evident from Figs. 9–11.

4. Conclusions

In this paper, a boundary layer analysis to study the combined effects of thermal radiation, viscous dissipation, Hall current on MHD convection, viscous, incompressible boundary layer fluid flow over a stretched vertical flat plate is presented. The governing partial differential equations of the free convective boundary layer flow are reduced to non-linear ordinary differential equations and solutions for primary velocity, secondary velocity and temperature field are obtained. The expressions for the skin-friction, heat transfer and mass transfer rates are also derived. In this study, only the assisting flow ($\lambda > 0$)

and forced convection flow ($\lambda = 0$) the thermal diffusivity decreases and these phenomena lead to the decreasing of energy ability that reduces the thermal boundary layer. It is found that the velocity, cross flow velocity and temperature distribution increases as the radiation parameter R or viscous dissipation parameter Ec increases.

Acknowledgments

The author is very thankful to the editor and reviewers for their encouraging comments and constructive suggestions to improve the presentation of this manuscript.

References

[1] Sakiadis BC. Boundary layer behaviour on continuous solid surfaces. AlChe J 1961;7:26–8.
[2] Banks WHH, Zaturska MB. Eigen solutions in boundary-layer flow adjacent to a stretching wall. IMA J Appl Math 1986;36:263–73.
[3] Sparrow EM, Abraham JP. Universal solutions for the stream wise variation of the temperature of a moving sheet in the presence of a moving fluid. Int J Heat Mass Transfer 2005;48:3047–56.
[4] Abraham JP, Sparrow EM. Friction drag resulting from the simultaneous imposed motions of a free stream and its bounding surface. Int J Heat Fluid Flow 2005;26:289–95.
[5] Lakshmisha KN, Venkateswaran S, Nath G. Three dimensional unsteady flow with heat and mass transfer over a continuous stretching surface. J Heat Trans 1988;110:590–5.
[6] Sakiadis BC. Boundary-layer behaviour on continuous solid surfaces: I. Boundary-layer equations for two dimensional and axisymmetric flow. AIChE J 1961;7:26–8.
[7] Crane LJ. Flow past a stretching plate. Z Angew Math Phys 1970;21:645–7.
[8] Sato H. The Hall effect in the viscous flow of ionized gas between parallel plates under transverse magnetic field. J Phys Soc Jpn 1961;16:1427–33.
[9] Sutton GW, Sherman A. Engineering magnetohydrodynamics. New York: McGraw-Hill; 1965.
[10] Pop I. The effect of Hall currents on hydromagnetic flow near an accelerated plate. J Math Phys Sci 1971;5:375–85.
[11] Raptis A, Ram PC. Effects of Hall current and rotation. Astrophys Space Sci 1984;106:257–64.
[12] Hossain MA, Rashid RIMA. Hall effect on hydromagnetic free convection flow along a porous flat plate with mass transfer. J Phys Soc Jpn 1987;56:97–104.
[13] Ezzat M, Zakaria M, Moursy M. Magnetohydrodynamic boundary layer flow past a stretching plate and heat transfer. J Appl Math 2004;1:9–21.
[14] Zakaria M. Magnetohydrodynamic viscoelastic boundary layer flow past a stretching plate and heat transfer. Appl Math Comput 2004;155:165–77.
[15] Yih KA. Free convection effect on MHD coupled heat and mass transfer of a moving permeable vertical surface. Int Commun Heat Mass Transfer 1999;26:95–104.
[16] Hossain MA. Effect of Hall current on unsteady hydromagnetic free convection flow near an infinite vertical porous plate. J Phys Soc Jpn 1986;55(7):2183–90.
[17] Hossain MA, Mahammad K. Effect of Hall current on hydromagnetic free convection flow near an accelerated porous plate. Jpn J Appl Phys 1988;27(8):1531–5.
[18] Salem AM, Abd El-Aziz M. Effect of Hall currents and chemical reaction on a hydromagnetic flow of a stretching vertical surface with internal heat generation/absorption. Appl Math Model 2008;32(7):1236–54.
[19] Ghosh SK, Anwar Bég O, Narahari M. Hall effects on MHD flow in a rotating system with heat transfer characteristics. Meccanica 2009;44:741–65.
[20] Abd El-Aziz M. Flow and heat transfer over an unsteady stretching surface with Hall effect. Meccanica 2010;45:97–109.
[21] Cess RD. The interaction of thermal radiation with free convection heat transfer. Int J Heat Mass Transfer 1966;9:1269–77.
[22] Arpaci VS. Effect of thermal radiation on the laminar free convection from a heated vertical plate. Int J Heat Mass Transfer 1968;11:871–81.
[23] Gnaneswara Reddy M, Bhaskar Reddy N. Radiation and mass transfer effects on unsteady MHD free convection flow past a vertical porous plate with viscous dissipation. Int J Appl Math Mech 2010;6(6):96–110.
[24] Gnaneswara Reddy M, Bhaskar Reddy N. Mass transfer and heat generation effects on MHD free convection flow past an inclined vertical surface in a porous medium. J Appl Fluid Mech 2011;4(3):7–11.
[25] Basiri Parsa A, Rashidi MM, Hayat T. MHD boundary-layer flow over a stretching surface with internal heat generation or absorption. Heat Trans – Asian Res 2013;42(6):500–14.
[26] Gnaneswara Reddy M. Effects of thermophoresis, viscous dissipation and joule heating on steady MHD heat and mass transfer flow over an inclined radiative isothermal permeable surface with variable thermal conductivity. Int J Heat Technol 2012;30(1):99–110.
[27] Cowling TG. Magnetohydrodynamics. New York: Interscience; 1957.
[28] Brewster MQ. Thermal radiative transfer and properties. New York: John Wiley & Sons; 1992.
[29] Jain MK, Iyengar SRK, Jain RK. Numerical methods for scientific and engineering computation. New Delhi: Wiley Eastern Ltd.; 1985.
[30] Ali FM, Nazar R, Arifin NM, Pop I. Effect of Hall current on MHD mixed convection boundary layer flow over a stretched vertical flat plate. Meccanica 2011;46:1103–12.



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